

Jenkins Task -1

What is Jenkins?

Jenkins is an open-source automation server used to build, test, and deploy applications automatically.

It is widely used in CI/CD pipelines to help developers deliver software faster, reliably, and with fewer errors.

Advantages of Jenkins

1. Automation – Automates build, test, and deployment processes, reducing manual work.
 2. CI/CD Support – Enables Continuous Integration and Continuous Delivery for faster releases.
 3. Extensible – Thousands of plugins available for Git, Docker, AWS, Maven, Kubernetes, etc.
 4. Open Source & Free – No licensing cost and strong community support.
 5. Easy Integration – Integrates easily with version control tools, cloud platforms, and DevOps tools.
 6. Scalable – Supports distributed builds using master–agent architecture.
 7. Improves Code Quality – Detects issues early through automated testing.
-

One-line interview answer

Jenkins is an open-source CI/CD tool that automates software build, testing, and deployment to enable faster and reliable application delivery.

- 1. Install Jenkins and run Jenkins on port number 8081.**

Steps:

Steps Followed to Change Jenkins Port to 8081

Jenkin Installation:

1. Added the Jenkins repository to the system:

2. `sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.repo`
3. Imported the Jenkins GPG key for package verification:
4. `sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key`
5. Installed Jenkins using the package manager:
6. `sudo dnf install jenkins -y`
7. Installed Java (Amazon Corretto) as Jenkins runtime dependency:
8. `sudo dnf install java-17-amazon-corretto -y`
9. Started and enabled Jenkins service:
10. `sudo systemctl start jenkins`
11. `sudo systemctl enable jenkins`

Steps to change port number:

1. Verified Jenkins was running on the default port and identified that the port is controlled at **service startup**, not inside Jenkins UI files.
2. Checked the Jenkins systemd service using `systemctl cat jenkins` to understand how Jenkins is started.
3. Created a **systemd override file** under `/etc/systemd/system/jenkins.service.d/override.conf`.
4. Set the environment variable `JENKINS_PORT=8081` inside the override file.
5. Reloaded systemd configuration using `systemctl daemon-reload`.
6. Restarted Jenkins service and verified Jenkins was listening on **port 8081** using `ss -tulnp`.
7. Allowed port **8081** in the EC2 security group and accessed Jenkins via browser.

✅ Why Override File Was Used Instead of Editing Service File

- The default Jenkins service file (`/usr/lib/systemd/system/jenkins.service`) is **managed by the package manager**.
- Editing the original service file can be **overwritten during Jenkins upgrades or OS updates**.
- Systemd overrides are the **recommended and safe approach** to customize service behavior.
- Overrides ensure **configuration persistence across reboots and updates**.
- This follows **Linux and DevOps best practices** for service management.

Installed java packages

```
[ec2-user@ip-172-31-21-117 ~]$ sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.repo
--2025-12-23 09:04:27-- https://pkg.jenkins.io/redhat-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.34.133, 2a04:4e42:78::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.34.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 192 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenkins.repo      100%[=====] 192 --.-KB/s   in 0s

2025-12-23 09:04:27 (18.5 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [192/192]

[ec2-user@ip-172-31-21-117 ~]$ sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
[ec2-user@ip-172-31-21-117 ~]$ sudo yum upgrade
Amazon Linux 2023 Kernel Livepatch repository                                249 kB/s | 29 kB   00:0
jenkins-stable                                                                13 kB/s | 833 B   00:0
jenkins-stable                                                                215 kB/s | 3.1 kB  00:0
Importing GPG key 0xE5975CA:
  userid      : "Jenkins Project <jenkinsci-board@googlegroups.com>"
  fingerprint: 6366 71E7 48BA 1F0A 08A0 9872 58A3 1057 E597 75CA
  from        : https://pkg.jenkins.io/redhat-stable/repodata/repomd.xml.key
is this ok [y/N]: y
jenkins-stable                                                                561 kB/s | 34 kB   00:0
Dependencies resolved.
Nothing to do.
Complete!
```

JENKINS INSTALATION ERROR:

1. When I was installing Jenkins from official documents java was not getting installed so tried with our institute documents as well still it was not installing

```
[ec2-user@ip-172-31-21-117 ~]$ sudo yum install fontconfig java-21-openjdk
Last metadata expiration check: 0:00:19 ago on Tue Dec 23 09:13:12 2025.
No match for argument: java-21-openjdk
Error: Unable to find a match: java-21-openjdk
[ec2-user@ip-172-31-21-117 ~]$ sudo wget -O /etc/yum.repos.d/jenkins.repo \
https://pkg.jenkins.io/redhat-stable/jenkins.repo

Last metadata expiration check: 0:01:48 ago on Tue Dec 23 09:13:12 2025.
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 0:01:49 ago on Tue Dec 23 09:13:12 2025.
No match for argument: java-21-openjdk
Error: Unable to find a match: java-21-openjdk
Last metadata expiration check: 0:01:50 ago on Tue Dec 23 09:13:12 2025.
Dependencies resolved.

complete!
[ec2-user@ip-172-31-21-117 ~]$ java -version
bash: java: command not found
[ec2-user@ip-172-31-21-117 ~]$ javac -version
bash: javac: command not found
[ec2-user@ip-172-31-21-117 ~]$ sudo yum install java-17-openjdk -y
Last metadata expiration check: 0:05:11 ago on Tue Dec 23 09:13:12 2025.
No match for argument: java-17-openjdk
Error: Unable to find a match: java-17-openjdk
[ec2-user@ip-172-31-21-117 ~]$ sudo dnf install java-17-amazon-corretto -y
Last metadata expiration check: 0:06:07 ago on Tue Dec 23 09:13:12 2025.
Dependencies resolved.
```

Then I have instaleed with following command “sudo yum install java-17-openjdk -y”

Reason:

- Your OS is **Amazon Linux 2023**.
- Amazon Linux 2023 **does not ship java-17-openjdk packages** in its repositories.
- Instead, AWS provides **Amazon Corretto**, which is:
 - AWS’s **official, supported OpenJDK distribution**
 - Optimized and maintained by AWS
 - Fully compatible with OpenJDK

Why dnf (not yum)

- Amazon Linux 2023 uses **dnf as the package manager**.
- yum commands may exist for compatibility, but **dnf is the native tool**.

Why Jenkins works with Corretto

- Jenkins requires **Java 11 or 17**.
- Amazon Corretto 17 meets Jenkins' Java requirement.
- That's why Jenkins starts successfully once Corretto is installed.

Simple one-line explanation (for docs/interview)

The command works because Amazon Linux 2023 uses Amazon Corretto as its supported Java distribution, which is available through dnf repositories and fully compatible with Jenkins.

If you want, I can also explain:

- Difference between **Amazon Linux 2 vs 2023**
- **Corretto vs OpenJDK**
- How to **set JAVA_HOME**
- Why Jenkins fails without Java

```
ec2-user@ip-172-31-21-117 ~]$ sudo dnf install java-17-amazon-corretto -y
Last metadata expiration check: 0:06:07 ago on Tue Dec 23 09:13:12 2025.
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:				
java-17-amazon-corretto	x86_64	1:17.0.17+10-1.amzn2023.1	amazonlinux	219 k
Installing dependencies:				
alsa-lib	x86_64	1.2.7.2-1.amzn2023.0.2	amazonlinux	504 k
cairo	x86_64	1.18.0-4.amzn2023.0.3	amazonlinux	717 k
dejavu-sans-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	1.3 M
dejavu-sans-mono-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	467 k
dejavu-serif-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	1.0 M
fontconfig	x86_64	2.15.94-2.amzn2023.0.2	amazonlinux	273 k
fonts-filesystem	noarch	1:12.0.5-12.amzn2023.0.2	amazonlinux	9.5 k
freetype	x86_64	2.13.2-5.amzn2023.0.1	amazonlinux	423 k
glib	x86_64	5.2.1-9.amzn2023.0.2	amazonlinux	48 k
google-noto-fonts-common	noarch	20240401-1.amzn2023.0.2	amazonlinux	17 k
google-noto-sans-vf-fonts	noarch	20240401-1.amzn2023.0.2	amazonlinux	593 k
harfbuzz	x86_64	1.3.14-3.amzn2023.0.3	amazonlinux	47 k

After installing Jenkins I have start and check the status Jenkins using systemctl start Jenkins and systemctl status Jenkins respectively

```
ec2-user@ip-172-31-21-117 ~]$ sudo systemctl start jenkins
ec2-user@ip-172-31-21-117 ~]$ sudo systemctl status jenkins
jenkins.service - Jenkins Continuous Integration Server
Loaded: loaded (/usr/lib/systemd/system/jenkins.service; disabled; preset: disabled)
Active: active (running) since Tue 2025-12-23 09:32:47 UTC; 11s ago
Main PID: 27501 (java)
Tasks: 44 (limit: 1115)
Memory: 382.0M
CPU: 14.928s
CGroup: /system.slice/jenkins.service
└─27501 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Dec 23 09:32:41 ip-172-31-21-117.ec2.internal jenkins[27501]: [LF]
Dec 23 09:32:41 ip-172-31-21-117.ec2.internal jenkins[27501]: [LF] *****
Dec 23 09:32:41 ip-172-31-21-117.ec2.internal jenkins[27501]: [LF] *****
Dec 23 09:32:47 ip-172-31-21-117.ec2.internal jenkins[27501]: 2025-12-23 09:32:47.648+0000 [id=30] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
Dec 23 09:32:47 ip-172-31-21-117.ec2.internal jenkins[27501]: 2025-12-23 09:32:47.671+0000 [id=23] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
Dec 23 09:32:47 ip-172-31-21-117.ec2.internal system[1]: Started jenkins.service - Jenkins Continuous Integration Server
Dec 23 09:32:47 ip-172-31-21-117.ec2.internal jenkins[27501]: 2025-12-23 09:32:47.938+0000 [id=47] INFO hudson.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven
Dec 23 09:32:47 ip-172-31-21-117.ec2.internal jenkins[27501]: 2025-12-23 09:32:47.939+0000 [id=47] INFO hudson.util.Retrier#start: Performed the action check updates server successfully at the at
Dec 23 09:32:52 ip-172-31-21-117.ec2.internal jenkins[27501]: 2025-12-23 09:32:52.742+0000 [id=69] WARNING hudson.DiskSpaceMonitorDescriptor#markNodeOfflineOnline: Making Built-in node offline to
```

Created directory and created a file to overwrite

```
ec2-user@ip-172-31-21-117 ~]$ sudo mkdir -p /etc/systemd/system/jenkins.service.d
ec2-user@ip-172-31-21-117 ~]$ sudo vi /etc/systemd/system/jenkins.service.d/override.conf
ec2-user@ip-172-31-21-117 ~]$ sudo systemctl daemon-reload
ec2-user@ip-172-31-21-117 ~]$ sudo systemctl restart jenkins
ec2-user@ip-172-31-21-117 ~]$ sudo systemctl cat jenkins | grep JENKINS_PORT
Environment="JENKINS_PORT=8080"
Environment="JENKINS_PORT=8081"
```

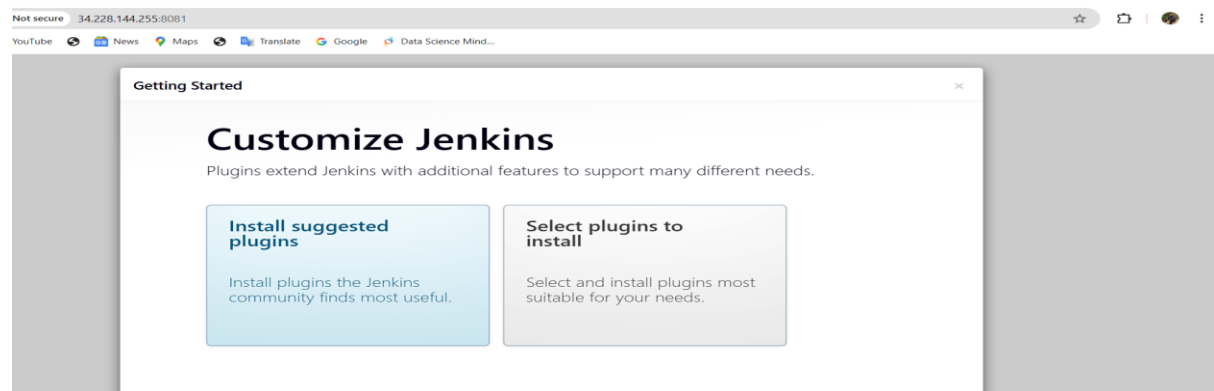
Jenkins port number overwrite

```
ec2-user@ip-172-31-21-117 ~]$ cat /etc/systemd/system/jenkins.service.d/override.conf
[Service]
Environment="JENKINS_PORT=8081"
```

Copied password from below path

```
cat: initialAdminPassword: No such file or directory
ec2-user@ip-172-31-21-117 bin]$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
9147b208e2ff4d759961755a380b72d1
ec2-user@ip-172-31-21-117 bin]$
```

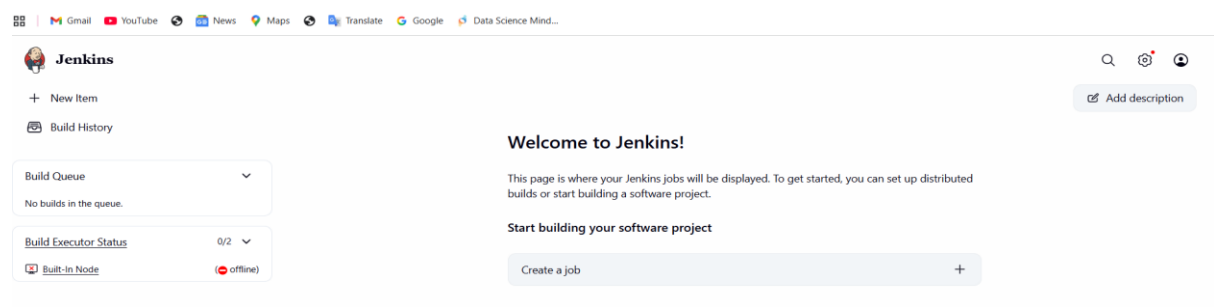
After entering password Jenkins opened, where I have selected suggested plugins



Jenkins URL



Setup Jenkins is done



2. Secure Jenkins server

Steps:

- I went to username "Arjumand Logo"
- Selected security
- Then changed the password, initially it was given as default.

Password

Password:

Confirm Password:

Session Termination

[Terminate All Sessions](#) ?

[Save](#) [Apply](#)

3. Create users called DevOps, Testing in Jenkins with Limited access.

Created an user called DevOps went to manage jenkins, create user and given credentials

Jenkins / Manage Jenkins / Jenkins' own user database / Create User

Create User

Username
DevOps

Password
.....

Confirm password
.....

Full name
DevOps

E-mail address
arjmand9794@gmail.com

[Create User](#)

Jenkins / Manage Jenkins / Security

Security Realm
Jenkins' own user database

☐ Allow users to sign up

Authorization
Project-based Matrix Authorization Strategy

	Overall	Credentials	Agent	Job	Run	View	SCM	Metrics
	Admin	Read	Write	Read	Read	Read	Read	Read
Anonymous	☐	☐	☐	☐	☐	☐	☐	☐
Authenticated Users	☐	☐	☐	☐	☐	☐	☐	☐
DevOps	☐	☒	☐	☐	☐	☐	☐	☐

[Save](#) [Apply](#)

User got limited access and created user in manage Jenkins after loggedin to user

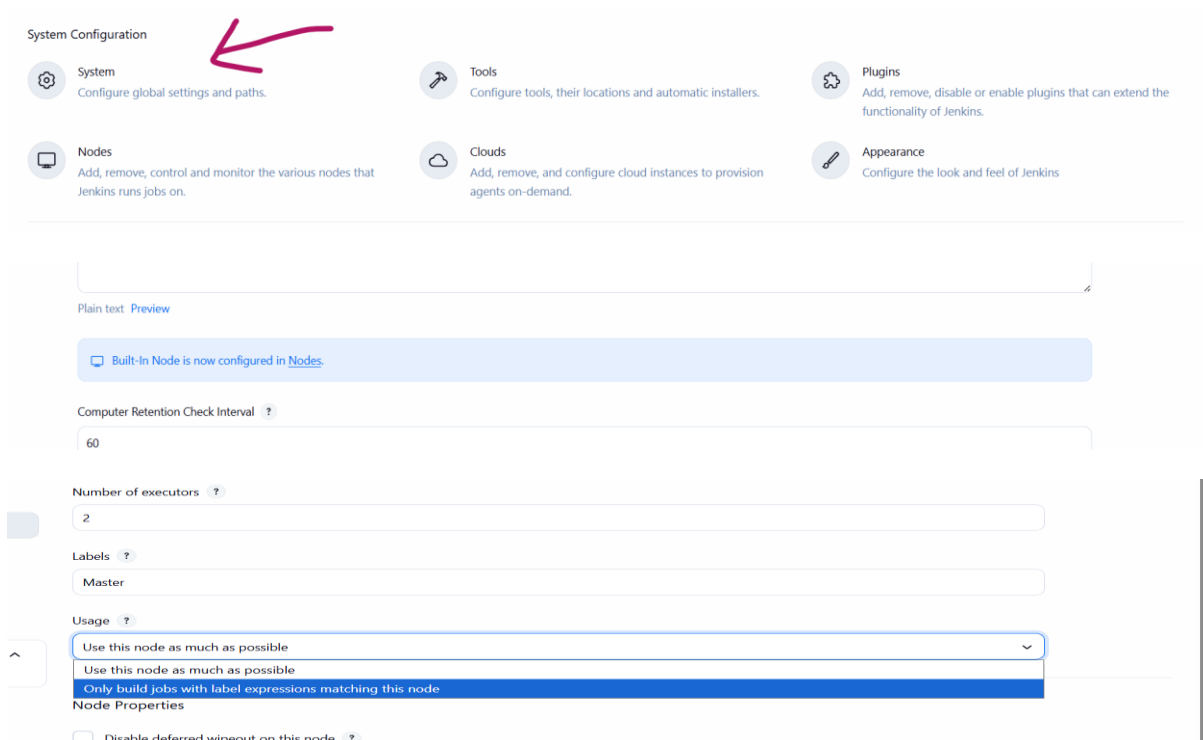
Jenkins / DevOps

[Add description](#)

DevOps Jenkins User ID: DevOps

- Profile
- Builds
- My Views
- Account
- Appearance
- Preferences
- Security
- Experiments
- Credentials

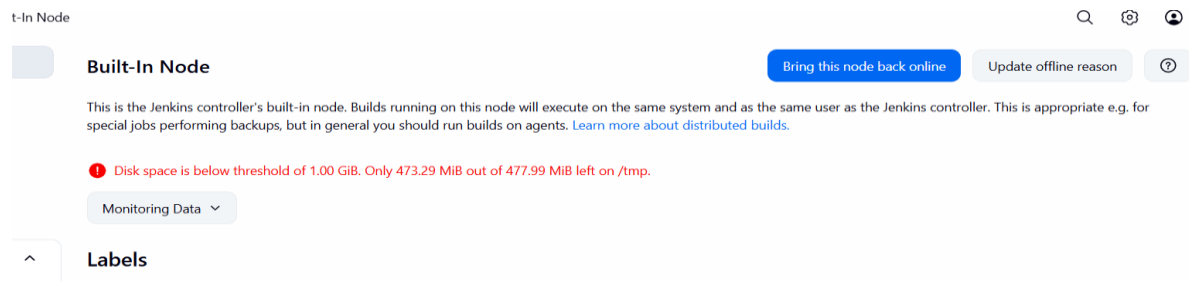
4. Configure labels and restrict the jobs to execute based on label only.



The screenshot shows the Jenkins 'System Configuration' page. A red arrow points to the 'System' option in the left sidebar. The main content area shows various configuration fields: 'Computer Retention Check Interval' (60), 'Number of executors' (2), 'Labels' (Master), and 'Usage' (Use this node as much as possible). The 'Usage' dropdown is open, showing three options: 'Use this node as much as possible', 'Use this node as much as possible', and 'Only build jobs with label expressions matching this node'. The 'Node Properties' section is also visible.

ERROR:

While I was creating master label I was facing issue related to storage,



The screenshot shows the Jenkins 't-In Node' page. The 'Built-In Node' section is active, showing a red error message: 'Disk space is below threshold of 1.00 GiB. Only 473.29 MiB out of 477.99 MiB left on /tmp.' The 'Monitoring Data' dropdown is set to 'Monitoring Data'. The 'Labels' section is also visible.

Steps followed for troubleshoot:

- Identified that the Jenkins built-in node went offline due to **low disk space threshold on /tmp**.
- Increased the EC2 **root EBS volume from 8 GB to 30 GB** and verified the filesystem size.
- Discovered **/tmp** was mounted as **tmpfs (RAM-based)**, causing Jenkins to still detect insufficient space.
- Redirected Jenkins temporary files to a **disk-based directory** using a systemd override and restarted Jenkins.
- Brought the node back online, verified node health, and confirmed the **label was created and usable**.

Here are 5 concise troubleshooting lines, including EBS modification and the key commands you used, ready for documentation:

- Detected Jenkins node offline due to low disk space on /tmp and verified usage using
- `df -h /tmp`
- Modified the EC2 **root EBS volume from 8 GB to 30 GB** via AWS Console and confirmed disk layout with
- `lsblk`
- Verified that /tmp was mounted as **tmpfs**, explaining the persistent disk warning:
- `df -h`
- Redirected Jenkins temporary files to disk by updating the systemd override and applied changes:
- `sudo vi /etc/systemd/system/jenkins.service.d/override.conf`
- `sudo systemctl daemon-reload`
- `sudo systemctl restart jenkins`
- Brought the Jenkins built-in node back online and validated successful recovery using the Jenkins UI.

Created label

Built-In Node

Mark this node temporarily offline ⓘ

This is the Jenkins controller's built-in node. Builds running on this node will execute on the same system and as the same user as the Jenkins controller. This is appropriate e.g. for special jobs performing backups, but in general you should run builds on agents. [Learn more about distributed builds.](#)

Monitoring Data ▾

Labels

Master

Projects tied to Built-In Node

None

Created first job in freestyle project

Enter an item name

» Required field

**Freestyle project**

This is the central feature of Jenkins. Jenkins will build your project, combining any SCM with any build system, and this can be even used for something other than software build.

**Pipeline**

Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows)

Selected option called restrict where can be run

☐ This project is parameterized ?
☐ Throttle builds ?
☐ Execute concurrent builds if necessary ?
☒ Restrict where this project can be run ?
 Label Expression ?

 Label Master matches 1 node. Permissions or other restrictions provided by plugins may further reduce that list.

Advanced ▾

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

⋮ Execute shell ?

Command

See [the list of available environment variables](#)

```
sleep 30
```

Now job will be executed only in Master label

Builds > ⋮

Today

⋮
#1
1:11 PM

✖
▾

Job is executed successfully

✓
#1 (Dec 23, 2025, 1:11:44 PM)

[Add description](#)
[Keep this build forever](#)

🕒
Started by user [Arjumand Mohiuddin](#)

Started 4 min 4 sec ago
Took [30 sec](#)

🕒
This run spent:

- 33 ms waiting;
- 30 sec build duration;
- 30 sec total from scheduled to completion.

</>
No changes.

5. Create Three sample jobs using the below

URL. <https://github.com/betawins/Techie horizon Login app.git>

Jenkins / All ▾ / New Item

New Item

Enter an item name

Select an item type

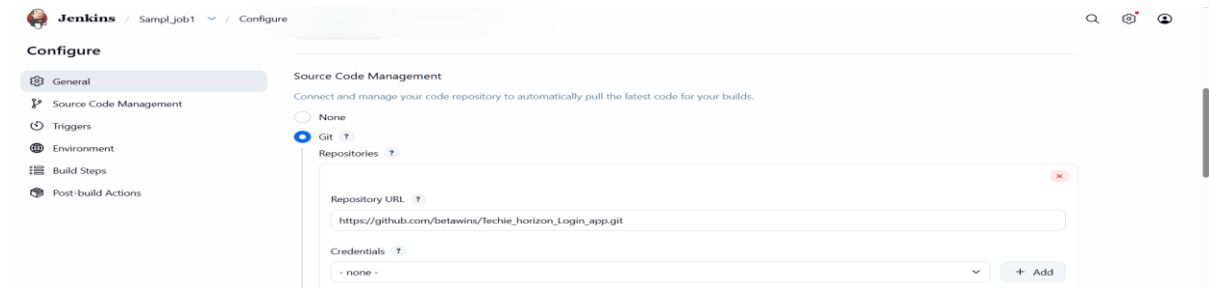
Freestyle project
 Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

```

Installed:
git-2.50.1-1.amzn2023.0.1.x86_64      git-core-2.50.1-1.amzn2023.0.1.x86_64      git-core-doc-2.50.1-1.amzn2023.0.1.noarch      perl-Error-1.0.17029-5.amzn2023.0.2.noarch
perl-File-Find-1.37-477.amzn2023.0.7.noarch      perl-git-2.50.1-1.amzn2023.0.1.noarch      perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64      perl-11b-0.65-477.amzn2023.0.7.x86_64

complete!
[ec2-user@ip-172-31-21-117 first_job]$ git --version
usage: git [-v] [--version] [-h] [--help] [-C <path>] [-c <name>=<value>]
           [--exec-path<path>] [--html-path] [--man-path] [--info-path]
           [-p | --paginate | -P | --no-pager] [--no-replace-objects] [--no-lazy-fetch]
           [--no-optional-locks] [--no-advice] [--bare] [--git-dir<path>]
           [--work-tree<path>] [--namespace<name>] [--config-env=<name>=<envvar>]
           [<command>] [<args>]
[ec2-user@ip-172-31-21-117 first_job]$ pwd
/var/lib/jenkins/jobs/first_job
[ec2-user@ip-172-31-21-117 first_job]$ cd ..
[ec2-user@ip-172-31-21-117 jobs]$ pwd
/var/lib/jenkins/jobs
[ec2-user@ip-172-31-21-117 jobs]$ cd /
[ec2-user@ip-172-31-21-117 jenkins]$ cd /
[ec2-user@ip-172-31-21-117 /]$ git --version
git version 2.50.1

```



Job is executed sample job 1 using URL without Master label

#1 (Dec 23, 2025, 1:48:11 PM)

Add description
Keep this build forever

Started by user [Arjumand Mohiuddin](#) (2 times)

Started 1 min 42 sec ago
Took 3.5 sec

This run spent:

- 7 min 28 sec waiting;
- 3.5 sec build duration;
- 7 min 32 sec total from scheduled to completion.

Revision: ef9c453d4993843ac49d37321316c2588f154495
Repository: https://github.com/betawins/Techie_horizon_Login_app.git

- refs/remotes/origin/master

No changes.

Second sample job using URL with Master label and confirm with status

Label Master matches 1 node. Permissions or other restrictions provided by plugins may further reduce that list.

Advanced

Source Code Management
Connect and manage your code repository to automatically pull the latest code for your builds.

None
Git

Repositories

Repository URL
https://github.com/betawins/Techie_horizon_Login_app.git

Credentials
none

Save
Apply

#1 (Dec 23, 2025, 1:57:59 PM)

Add description
Keep this build forever

Started by user [Arjumand Mohiuddin](#)

Started 1 min 2 sec ago
Took 0.21 sec

This run spent:

- 4 ms waiting;
- 0.21 sec build duration;
- 0.22 sec total from scheduled to completion.

Revision: ef9c453d4993843ac49d37321316c2588f154495
Repository: https://github.com/betawins/Techie_horizon_Login_app.git

- refs/remotes/origin/master

No changes.

New job with different label hotfix which is not there

New Item

Enter an item name

Select an item type

**Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

**Pipeline**
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.

☒ Restrict where this project can be run ?

Label Expression ?

Hotfix

⚠ No agent/cloud matches this label expression. Did you mean 'Master' instead of 'Hotfix'?

Advanced ▾

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/betawins/Techie_horizon_Login_app.git

As you can see my third sample is not executed yet because that label is not there

+ New Item

Started by user Arjumand Mohiuddin
Started by user Arjumand Mohiuddin
'jenkins' doesn't have label 'Hotfix'
Waiting for 42 sec

sample_job3 ▾

Build Executor Status
(0 of 2 executors busy)

All +

S	W	Name ↓	Last Success	Last Failure	Last Duration	
✓	☀	first_job	51 min #1	N/A	30 sec	▶
✓	☀	Sampl_job1	15 min #1	N/A	3.5 sec	▶
✓	☀	Sample_job2	5 min 32 sec #1	N/A	0.21 sec	▶
⋮	☀	sample_job3	N/A	N/A	N/A	▶

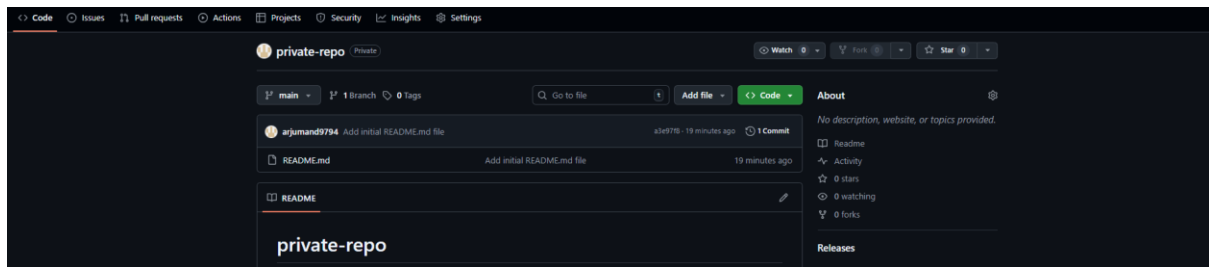
Add description

6. Create one Jenkins job using git hub Private repository.

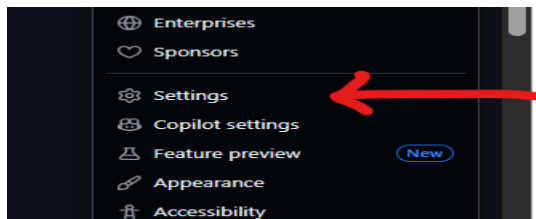
Steps:

1. Create a Private Repository in GitHub

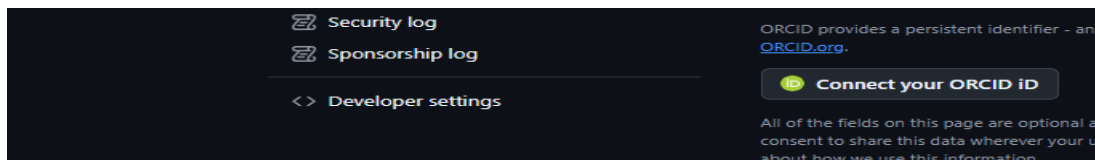
- Logged in to GitHub.
- Created a **private repository** named private-repo.
- Added a README.md file (initial commit).
- Noted the **default branch name: main**.
- Copied the **HTTPS repository URL**:
- <https://github.com/arjumand9794/private-repo.git>



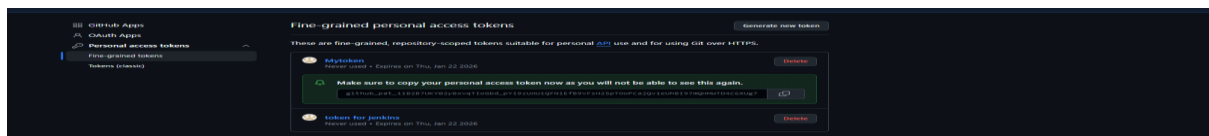
- Go to logo and the settings



- Go to developer setting



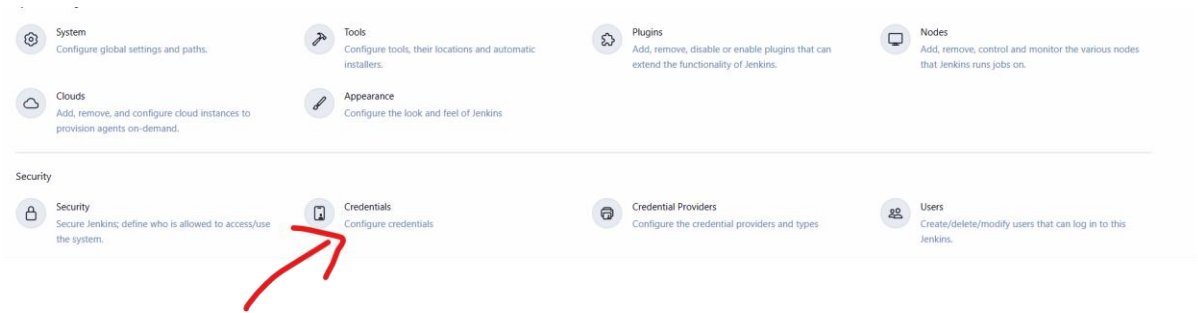
- Create a token classic



2. Create GitHub Credentials in Jenkins

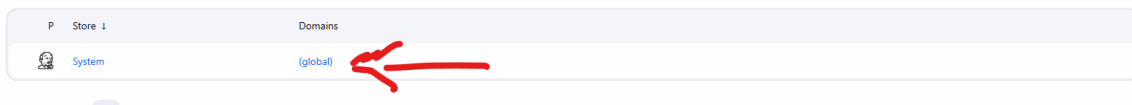
- Went to Jenkins Dashboard → Manage Jenkins → Credentials.
- Selected (global) → Add Credentials.
- Chose:
 - Kind: Username with password
 - Username: GitHub username
 - Password: GitHub Personal Access Token (PAT)
 - ID: github-private-repo (simple name, no URL)
 - Description: GitHub private repository access
- Saved the credentials.

- Go to credentials

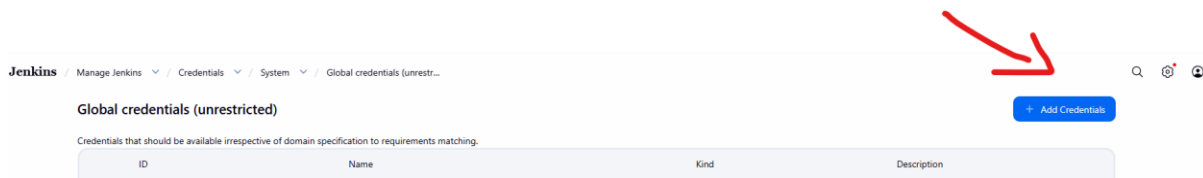


- Go to global

Stores scoped to Jenkins



- Go to add credentials



Add user name, password (token) Id then add discription and then save

Kind: Username with password

Scope: Global (Jenkins, nodes, items, all child items, etc)

Username:

☐ Treat username as secret

Password:

ID:

Description:

3. Create a New Jenkins Job

- Clicked New Item in Jenkins.
- Entered job name: github-private-repo.
- Selected Freestyle project.
- Clicked OK.

New Item

Enter an item name

github-private-repo

Select an item type



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Configure Git Repository in the Job

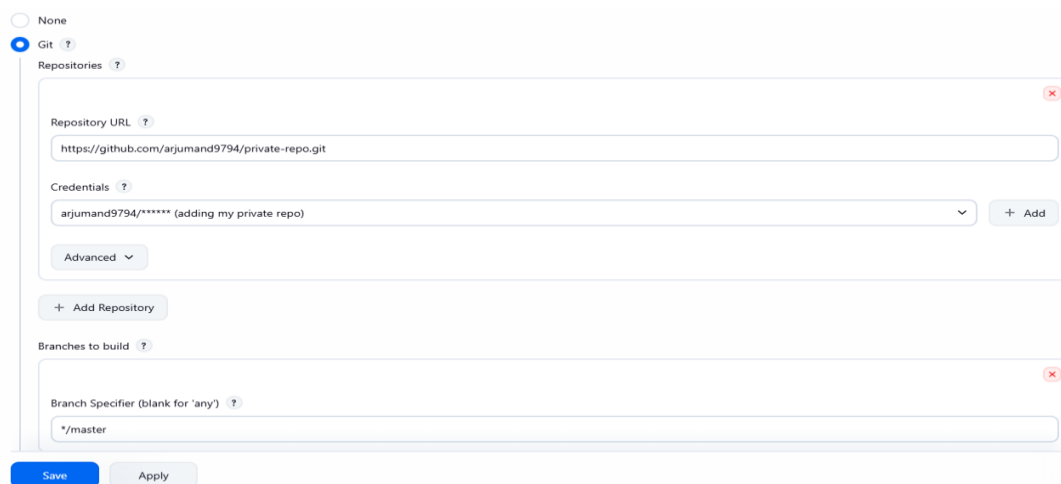
- Opened Configure for the job.
- Under Source Code Management, selected Git.
- Pasted the private repository URL:
- <https://github.com/arjumand9794/private-repo.git>
- Selected the saved GitHub credentials.

5. Configure Correct Branch (IMPORTANT STEP)

- Under **Branches to build**, set:
 - `*/main`
 - This was required because the repository uses **main**, not master.
-

6. Save and Build the Job

- Clicked **Save**.
- Clicked **Build Now**.



The screenshot shows a configuration window for a Git repository. At the top, there are radio buttons for 'None' and 'Git', with 'Git' selected. Below this is a section titled 'Repositories' with a search icon. It contains a 'Repository URL' field with the value 'https://github.com/arjumand9794/private-repo.git' and a 'Credentials' dropdown menu showing 'arjumand9794/* (adding my private repo)' with an 'Add' button. An 'Advanced' dropdown is also present. Below the 'Repositories' section is a '+ Add Repository' button. The next section is 'Branches to build' with a search icon. It contains a 'Branch Specifier (blank for \'any\')' field with the value '*/master'. At the bottom, there are 'Save' and 'Apply' buttons.

Verify Build Execution

- Opened Build #2.
- Checked:
 - Status: Green tick (SUCCESS)
 - Console Output: No errors

- Git Build Data:
 - Branch: refs/remotes/origin/main
 - Commit ID displayed
- Confirmed the job executed successfully.

✓ Final Result

The screenshot displays the Jenkins interface for a job named 'github-private-repo'. The job is in a successful state, indicated by a green checkmark and the label '#2 (Dec 23, 2025, 4:24:43 PM)'. The left sidebar contains navigation links: Status, Changes, Console Output, Edit Build Information, Delete build *#2, Timings, Git Build Data, and Previous Build. The main content area shows the build details, including the user 'Arjumand Mohiuddin' and the Git build data. The Git build data section shows the revision 'a3e97f84ce508f14e07d7464fedbb6b3fced3dc' and the repository 'https://github.com/arjumand934/private-repo.git'. The branch is 'refs/remotes/origin/main'. The build took 0.25 seconds to complete.

Jenkins / github-private-repo / #2

Status #2 (Dec 23, 2025, 4:24:43 PM)

Started by user Arjumand Mohiuddin

This run spent:

- 2 ms waiting;
- 0.26 sec build duration;
- 0.26 sec total from scheduled to completion.

git

Revision: a3e97f84ce508f14e07d7464fedbb6b3fced3dc

Repository: <https://github.com/arjumand934/private-repo.git>

- refs/remotes/origin/main

No changes.

Add description Keep this build forever

Started 1 min 19 sec ago Took 0.25 sec